PHENIKAA UNIVERSITY
SCHOOL OF INFORMATION SYSTEMS

COURSE CSE703095
SOFTWARE REQUIREMENTS

SAMPLE REPORT – LAB 05

Modeling Techniques & Notations

Topic: Illustrative example of a completed report (reference only, do not copy)

Case study project: MedBook – Online Medical Appointment Booking System
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1 Mapping State-transition Events to Use Cases

Event	State transition	Related use case
Receptionist confirms	pending_confirmation → confirmed	UC-05
Doctor completes visit	confirmed → completed	UC-07
Patient cancels (>2h)	(*) → cancelled	UC-04

2 DFD Level 2 – ”Drug Safety Warning” Process (hand-drawn)

Process 5.0 is decomposed into 3 sub-processes: 5.1 Look up patient allergy list (reads store D4: Allergy List), 5.2 Look up drug interaction catalog (reads store D5: Interaction Catalog), 5.3 Match & generate warning (writes warning log D6). This level of decomposition is detailed enough to hand off to the software design team.

3 State Machine vs. DFD Comparison for NFR-007

The state machine describes *when* an appointment can transition between states, but **cannot** show *where* the `unique(doctor, slot)` constraint check happens in the system. Conversely, DFD level 1 (process 2.0 ”Book appointment” writing to store D2 ”Appointments”) clearly shows where conflict-checking logic must sit – so **DFD is better suited** for specifying *where* NFR-007 is implemented, while the state machine is better suited for specifying the *consequence* of conflict detection (the appointment is not created, no new state is reached).

4 Extension Code: DFD Level 2 in Code

```
DFD_LEVEL2_SAFETY = r"""
\begin{tikzpicture}[proc/.style={draw, circle, fill=penilight}]
\node[proc] (p51) at (0,0) {5.1 Look up\\allergies};
\node[proc] (p52) at (4,0) {5.2 Look up\\interactions};
\node[proc] (p53) at (8,0) {5.3 Match \&\\warn};
\draw[->] (p51) -- (p53);
\draw[->] (p52) -- (p53);
\end{tikzpicture}
"""
```

5 Added Guard Condition

The transition edge `pending_confirmation` → `confirmed` was given a guard: `[slot_available == True AND deposit_paid == valid]`.

6 Self-assessment

Criterion	Max	Self-score	Notes
Complete state machine	2.5	2.5	
Correct DFD 0/1/2 notation	3.5	3.0	DFD level 2 needs tighter data-flow balance
Comparative analysis	2.0	2.0	
Presentation	2.0	2.0	
Total	**10.0**	**9.5**	